Yakeen Plus (2025)

PRACTICE TEST- 01

DURATION :200 Minutes | DATE : 15/09/2024 | M.MARKS :720

ANSWER KEY

PHYSICS		CHEMISTRY		BOTANY		ZOOLOGY	
1.	(2)	51.	(2)	101.	(1)	151.	(3)
2.	(2)	52.	(4)	102.	(2)	152.	(4)
3.	(2)	53.	(4)	103.	(4)	153.	(4)
4.	(4)	54.	(4)	104.	(1)	154.	(1)
5.	(4)	55.	(1)	105.	(3)	155.	(1)
6.	(3)	56.	(2)	106.	(2)	156.	(3)
7.	(4)	57.	(1)	107.	(3)	157.	(3)
8.	(4)	58.	(4)	108.	(2)	158.	(4)
9.	(2)	59.	(3)	109.	(2)	159.	(1)
10.	(3)	60.	(4)	110.	(2)	160.	(3)
11.	(3)	61.	(1)	111.	(4)	161.	(1)
12.	(1)	62.	(4)	112.	(1)	162.	(3)
13.	(2)	63.	(3)	113.	(1)	163.	(4)
14.	(3)	64.	(2)	114.	(2)	164.	(3)
15.	(4)	65.	(4)	115.	(3)	165.	(3)
16.	(3)	66.	(3)	116.	(4)	166.	(2)
17.	(2)	67.	(1)	117.	(2)	167.	(1)
18.	(2)	68.	(2)	118.	(4)	168.	(2)
19.	(2)	69.	(4)	119.	(3)	169.	(3)
20.	(4)	70.	(4)	120.	(2)	170.	(2)
21.	(4)	71.	(1)	121.	(1)	171.	(3)
22.	(3)	72.	(1)	122.	(3)	172.	(3)
23.	(3)	73.	(1)	123.	(4)	173.	(1)
24.	(1)	74.	(3)	124.	(2)	174.	(4)
25.	(3)	75.	(3)	125.	(1)	175.	(2)
26.	(1)	76.	(3)	126.	(4)	176.	(3)
27.	(1)	77.	(2)	127.	(3)	177.	(2)
28.	(4)	78.	(1)	128.	(4)	178.	(4)
29.	(2)	79.	(1)	129.	(2)	179.	(1)
30.	(1)	80.	(1)	130.	(3)	180.	(1)
31.	(2)	81.	(1)	131.	(2)	181.	(2)
32.	(3)	82.	(1)	132.	(3)	182.	(3)
33.	(1)	83.	(1)	133.	(1)	183.	(1)
34.	(3)	84.	(2)	134.	(1)	184.	(3)
35.	(3)	85.	(1)	135.	(3)	185.	(4)
36.	(2)	86.	(1)	136.	(2)	186.	(3)
37.	(3)	87.	(1)	137.	(1)	187.	(4)
38.	(3)	88.	(1)	138.	(2)	188.	(1)
39.	(2)	89.	(2)	139.	(4)	189.	(4)
40.	(2)	90.	(1)	140.	(3)	190.	(4)
41.	(3)	91.	(2)	141.	(4)	191.	(1)
42.	(1)	92.	(2)	142.	(4)	192.	(4)
43.	(4)	93.	(4)	143.	(3)	193.	(4)
44.	(3)	94.	(4)	144.	(3)	194.	(3)
45.	(4)	95.	(1)	145.	(3)	195.	(4)
46.	(3)	96.	(1)	146.	(3)	196.	(2)
47.	(3)	97.	(3)	147.	(2)	197.	(3)
48.	(3)	98.	(2)	148.	(4)	198.	(4)
49.	(4)	99.	(3)	149.	(2)	199.	(3)
50.	(3)	100.	(3)	150.	(4)	200.	(3)

SECTION-(I) PHYSICS

1. **(2)**

$$P = \frac{a - t^2}{bx}$$

$$P = \left[ML^{-1}T^{-2}\right]$$

a will have same unit as t^2. So, $a = [T^2]$

$$b = \left[\frac{a - t^2}{Px}\right] = \frac{\left[T^2\right]}{\left[ML^{-1}T^{-2}L\right]} = \left[M^{-1}L^0T^4\right]$$

$$\frac{a}{b} = \frac{\left[T^2\right]}{\left[M^{-1}L^0T^4\right]} = \left[ML^0T^{-2}\right]$$

[New NCERT Class 11th Page No. 8]

2. **(2)**

$F = kA^x v^y \rho^z$

$$\Rightarrow \quad MLT^{-2} = \left[L^2\right]^x \left[LT^{-1}\right]^y \left[ML^{-3}\right]^z$$

$\Rightarrow \quad z = 1, 2x + y - 3z = 1, -y = -2 \Rightarrow y = 2$

$\Rightarrow \quad x = 1$

$$F = kA^1 v^2 \rho^1 = Av^2\rho, k = 1$$

[New NCERT Class 11th Page No. 9]

3. **(2)**

$$v = at + \frac{b}{t + c}$$

By the principle of homogeneity,

$$c = t = [T]$$

$$at = v \Rightarrow a = \left[LT^{-2}\right]$$

$$\frac{b}{T} = LT^{-1} \Rightarrow b = [L]$$

[New NCERT Class 11th Page No. 8]

4. **(4)**

$$P = P_0 e^{\left(-at^2\right)}$$

The power of exponent is dimensionless,

$$at^2 = \left[M^0L^0T^0\right]$$

$$\boxed{\alpha = \left[T^{-2}\right]}$$

[New NCERT Class 11th Page No. 7]

5. **(4)**

$n_1u_1 = n_2u_2$

$$\Rightarrow \quad \frac{4g}{cm^3} = n_2 \times \frac{100g}{10^3 cm^3}$$

$$\Rightarrow \quad \boxed{n_2 = 40}$$

[New NCERT Class 11th Page No. 8]

6. **(3)**

P-Q, no subtraction is there because of different dimension.

[New NCERT Class 11th Page No. 7]

7. **(4)**

Relation between unit and magnitude

(nu = constant).

[New NCERT Class 11th Page No. 3]

8. **(4)**

A dimensionally incorrect equation may be correct.

[New NCERT Class 11th Page No. 8]

9. **(2)**

$P \rightarrow$ Pressure

$V \rightarrow$ Volume

$T \rightarrow$ Temperature

$R \rightarrow$ const

$$P = \frac{a}{V^2}$$

$$ML^{-1}T^{-2} = \frac{a}{(L^3)^2}$$

$a = ML^{-1}T^{-2}L^6 = [M^1L^5T^{-2}]$

[New NCERT Class 11th Page No. 9]

10. **(3)**

Displacement vector $\vec{r} = \Delta x\hat{i} + \Delta y\hat{j} + \Delta z\hat{k}$

$$= (3-2)\hat{i} + (4-3)\hat{j} + (5-5)\hat{k} = \hat{i} + \hat{j}$$

[New NCERT Class 11th Page No. 8]

11. **(3)**

$$\vec{R} = \hat{i} + \hat{j} + \sqrt{2}\hat{k}$$

Comparing the given vector with $\vec{R} = R_x\hat{i} + R_y\hat{j} + R_z\hat{k}$

$$R_x = 1, R_y = 1, R_z = \sqrt{2}$$

and $\left|\vec{R}\right| = \sqrt{R_x^2 + R_y^2 + R_z^2} = 2$

$$\cos\alpha = \frac{R_x}{R} = \frac{1}{2} \Rightarrow \alpha = 60°$$

$$\cos\beta = \frac{R_y}{R} = \frac{1}{2} \Rightarrow \beta = 60°$$

$$\cos\gamma = \frac{R_z}{R} = \frac{1}{\sqrt{2}} \Rightarrow \gamma = 45°$$

[New NCERT Class 11th Page No. 29

12. (1)

$(0.3)^2 + (-0.4)^2 + C^2 = 1$

$C = \sqrt{0.75}$

[New NCERT Class 11th Page No. 30]

13. (2)

$\vec{A}.(\vec{B} \times \vec{A}) = 0$

[New NCERT Class 11th Page No. 33]

14. (3)

$A + B = 30° + 45° = 75°$

[New NCERT Class 11th Page No. 32]

15. (4)

$|\vec{A} \times \vec{B}| = \sqrt{3}\vec{A}.\vec{B}$

$AB\sin\theta = \sqrt{3}AB\cos\theta$

$\tan\theta = \sqrt{3}$

$\theta = 60°$, then

$R = (A^2 + B^2 + 2AB\cos 60°)^{1/2}$

$= (A^2 + B^2 + AB)^{1/2}$

[New NCERT Class 11th Page No. 33]

16. (3)

$\vec{C} = \vec{A} + \vec{B}$

$|\vec{C}| = |\vec{A} + \vec{B}|$

$C^2 = A^2 + B^2 + 2AB\cos\theta$

$13^2 = 12^2 + 5^2 + 2 \times 12 \times 5\cos\theta$

$169 = 144 + 25 + 120\cos\theta$

$\theta = \frac{\pi}{2}$

[New NCERT Class 11th Page No. 30]

17. (2)

$F = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$

[New NCERT Class 11th Page No. 32]

18. (2)

Let $\hat{n}_1$ and $\hat{n}_2$ are the two unit vectors, then the sum is

$\vec{n}_s = \hat{n}_1 + \hat{n}_2$ or $n_s^2 = n_1^2 + n_2^2 + 2n_1n_2\cos\theta$

Since it is given that n_s is also a unit vector, therefore

$1 = 1 + 1 + 2\cos\theta \Rightarrow \cos\theta = -\frac{1}{2} \therefore \theta = 120°$

Now the difference vector is

$\vec{n}_d = \hat{n}_1 - \hat{n}_2$ or $n_d^2 = n_1^2 + n_2^2 - 2n_1n_2\cos\theta$

$= 1 + 1 - 2\cos 120°$

$\therefore n_d^2 = 2 - 2\left(-\frac{1}{2}\right) = 2 + 1 = 3$

$\Rightarrow n_d = \sqrt{3}$

[New NCERT Class 11th Page No. 33]

19. (2)

$x^2 + 7x + 12 = 0$

$(x + 3)(x + 4) = 0$

$x = -3, -4$

[New NCERT Class 11th Page No. 3]

20. (4)

$S_\infty = \frac{a}{1-r}$

$= \frac{1}{1-1/4} = \frac{4}{3}$

[New NCERT Class 11th Page No. 2]

21. (4)

$\log_a a = 1$

$\log_{10} 10 = 1$

[New NCERT Class 11th Page No. 3]

22. (3)

$P^2 = 2m$ E.

$m \propto P^2$

[New NCERT Class 11th Page No. 3]

23. (3)

$\int_0^1 (x^3 + 1)dx$

$= \left(\frac{x^4}{4} + x\right)_0^1 = \frac{1}{4} + 1 = \frac{5}{4}$

[New NCERT Class 11th Page No. 2]

24. (1)

$P + Q = 18$

Also $P^2 + Q^2 + 2PQ\cos\theta = (12)^2 = 144$

Also $\frac{Q\sin\theta}{P + Q\cos\theta} = \tan 90° = \infty$

Therefore, $P + Q\cos\theta = 0$

Solving them, we find their magnitudes as 5 and 13.

[New NCERT Class 11th Page No. 33]

25. (3)

By traingle law

[New NCERT Class 11th Page No. 32]

26. **(1)**

$$\eta = \frac{F}{av} = \frac{[MLT^{-2}]}{[L][LT^{-1}]} = [ML^{-1}T^{-1}]$$

[New NCERT Class 11th Page No. 8]

27. **(1)**

On changing the units of measurement of a physical quantity, the numerical value either decreases or increases.

If the system of measurement is large, then the numerical value will be smaller if the system of measurement is smaller, then numerical value will be higher.

[New NCERT Class 11th Page No. 10]

28. **(4)**

$\vec{D} = \vec{A} - \vec{B}$

$|\vec{D}|^2 = A^2 + B^2 - 2AB\cos\theta$

$A^2 + B^2 = A^2 + B^2 - 2AB\cos\theta$

$\cos\theta = 0$

$\boxed{\theta = 90^\circ}$

[New NCERT Class 11th Page No. 30]

29. **(2)**

$R_{max} = |\vec{A} + \vec{B}|$

and $R_{min} = |\vec{A} - \vec{B}|$

$|\vec{A} + \vec{B}| \leqslant R \leqslant |\vec{A} - \vec{B}|$

[New NCERT Class 11th Page No. 31]

30. **(1)**

$y = x^2 - 2x + \sin x$

$$\frac{dy}{dx} = 2x - 2 + \cos x$$

[New NCERT Class 11th Page No. 7]

31. **(2)**

Surface tension	Kg s^{-2}
Pressure	Kg m^{-1} s^{-2}
Viscosity	Kg m^{-1} s^{-1}
Impulse	Kg m s^{-1}

[New NCERT Class 11th Page No. 5]

32. **(3)**

Work: $w = Fd$

$[ML^2T^{-2}]$

Torque:-

$\tau = Fr$

$[ML^2T^{-2}]$

[New NCERT Class 11th Page No. 7]

33. **(1)**

Astronomical unit, light year and parsec, each is used to measure large distance in our universe.

1 AU = 1.5×10^{11} m

1 ly = 9.46×10^5 m

1 per sec = 3.07×10^{16} m

[New NCERT Class 11th Page No. 5]

34. **(3)**

$(1 + x)^3 = 1 + 3x + 3x^2 + x^3$

$x << 1,$

$= 1 + 3x$

[New NCERT Class 11th Page No. 4]

35. **(3)**

$$S_n = \frac{n}{2}\left(29 + (n-1)d\right)$$

$$S_n = \frac{n}{2}\left(1^{st}\text{ term} + n^{th}\text{ term}\right)$$

[New NCERT Class 11th Page No. 33]

36. **(2)**

$$KE = \frac{P^2}{2m}$$

$\boxed{KE \propto P^2}$

"graph will be parabolic'.

[New NCERT Class 11th Page No. 18]

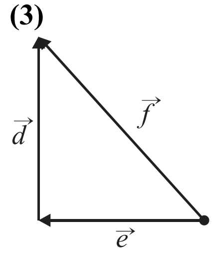
37. **(3)**

$y = \sin 4t$

$\boxed{\frac{dy}{dt} = 4\cos 4t}$

[New NCERT Class 11th Page No. 30]

38. **(3)**



$\vec{d} + \vec{e} = \vec{f}$

[New NCERT Class 11th Page No. 32]

39. **(2)**

$\vec{A}.\vec{B} = 0$

$\left(2\hat{i} + 3\hat{j} + 8\hat{k}\right).\left(4\hat{i} - 4\hat{j} + 2\hat{k}\right) = 0$

$8 - 12 + 8\propto = D$

$8\propto = 4$

$\boxed{\propto = 1/2}$

[New NCERT Class 11th Page No. 33]

40. (2)

$|\vec{A}+\vec{B}| = |\vec{A}| = |\vec{B}|$

$|\vec{A}+\vec{B}| = \sqrt{A^2+B^2+2AB\cos\theta}$

$A^2 = A^2 + A^2 + 2A^2\cos\theta$

$-\frac{1}{2} = \cos\theta \Rightarrow \theta = 120°$

[New NCERT Class 11th Page No. 31]

41. (3)

$$v = w\times r\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

$= -2\hat{i} + 2\hat{k}$

[New NCERT Class 11th Page No. 41]

42. (1)

Area $= \frac{1}{2}|\vec{A}\times\vec{B}|$

[New NCERT Class 11th Page No. 34]

43. (4)

$\alpha = \frac{F}{v^2}\sin(\beta t)$

dimensionless dimensionless

So $\alpha = \frac{[F]}{[v^2]} = \frac{[M^1L^1T^{-2}]}{[L^1T^{-1}]^2} = M^1L^{-1}T^0$

$\beta = \frac{1}{[t]} = \frac{1}{[T]} = M^0L^0T^{-1}$

[New NCERT Class 11th Page No. 8]

44. (3)

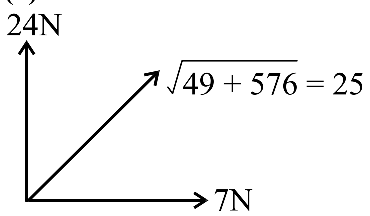
$\sin^2\alpha + \sin^2\beta + \sin\gamma$

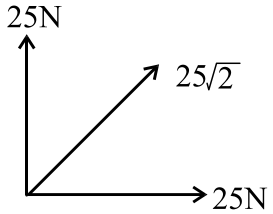
$= 1-\cos^2\alpha + 1-\cos^2\beta + 1-\cos^2\gamma$

$= 3-\left(\cos^2\alpha+\cos^2\beta+\cos^2\gamma\right) = 3-1 = 2$

[New NCERT Class 11th Page No. 33]

45. (4)





$F = 25\sqrt{2}$

[New NCERT Class 11th Page No. 33]

46. (3)

A dimensionaless quantity may have a unit.

Example : Angle unit radian.

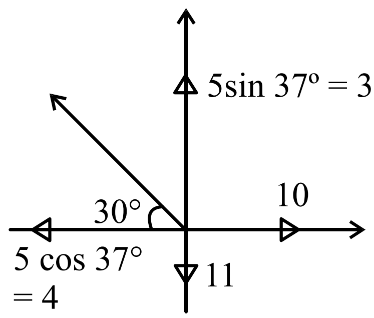
[New NCERT Class 11th Page No. 8]

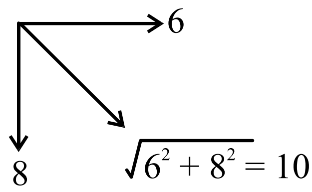
47. (3)

A unit less quantity never has a non zero dimension.

[New NCERT Class 11th Page No. 8]

48. (3)





Resultant = 10

[New NCERT Class 11th Page No. 33]

49. (4)

By putting the dimensions of each quantity both the sides we get $[T^{-1}] = [M]^x[MT^{-2}]^y$

Now, comparing the dimensions of quantities in both sides we get $x+y=0$ and $2y=1$

$\therefore\ x = -\frac{1}{2},\ y = \frac{1}{2}$

[New NCERT Class 11th Page No. 7]

50. (3)

Let $v = k\,g^y\lambda^z\rho^\delta$. Now by substituting the dimensions of each quantities and equating the powers of M, L and T we get $\delta = 0$ and $y = \frac{1}{2},\ z = \frac{1}{2}$.

[New NCERT Class 11th Page No. 8]

SECTION-(II) CHEMISTRY

51. (2)

Moles = $\dfrac{\text{Given mass}}{\text{Molar mass}}$

Moles of D_2O = $\dfrac{5}{20} = \dfrac{1}{4}$

Number of neutrons = $10 \times \dfrac{1}{4} \times N_A = 2.5\ N_A$

[New NCERT Class 11th Page No. 5]

52. (4)

The mass of one mole of carbon will change.

[New NCERT Class 11th Page No. 3]

53. (4)

- An element of a substance contains only one kind of atoms.
- A compound can be decomposed into its components.
- All homogeneous mixtures are called as solutions.

[New NCERT Class 11th Page No. 10]

54. (4)

Mass of 1 molecule of water = 18 amu

$= 18 \times 1.66 \times 10^{-24}$ g

Volume of 1 molecule = $\dfrac{\text{mass}}{\text{density}} = \dfrac{29.88 \times 10^{-24}}{1}$

$= 29.88 \times 10^{-24} = 2.988 \times 10^{-23}$ mL

[New NCERT Class 11th Page No. 15]

55. (1)

Let mole % of ^{26}Mg be x

$\therefore \dfrac{(21-x)25 + x(26) + 79(24)}{100} = 24.31$

x = 10%

[New NCERT Class 11th Page No. 6]

56. (2)

The law of definite proportion states that given chemical compound always contains the same elements in the exact same proportion by mass.

[New NCERT Class 11th Page No. 10]

57. (1)

The modern atomic weight scale is based on C^{12} isotope.

[New NCERT Class 11th Page No. 16]

58. (4)

Sulphur dioxide and sulphur trioxide follow law of multiple proportion.

[New NCERT Class 11th Page No. 7]

59. (3)

% of O = $\dfrac{16 \times 27}{(100 + 3 \times 310)} \times 100 = 41.94\%$

[New NCERT Class 11th Page No. 20]

60. (4)

$$\begin{array}{ccc} X & : & Y \\ \dfrac{75.8}{75} & & \dfrac{24.2}{16} \\ 1 & : & 3/2 \\ 2 & : & 3 \end{array} \Rightarrow X_2Y_3$$

[New NCERT Class 11th Page No. 18]

61. (1)

1 amu = $\dfrac{1}{12}$ mass of C–12 isotope.

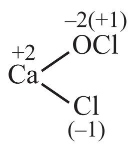
[New NCERT Class 11th Page No. 8]

62. (4)

m = $\dfrac{1000\ M}{1000\rho - MM_1}$

[New NCERT Class 11th Page No. 17]

63. (3)



[New NCERT Class 11th Page No. 14]

64. (2)

Equivalent weight of tribasic acid

$= \frac{\text{Molecular mass}}{\text{Basicity}} = \frac{W}{3}$

[New NCERT Class 11th Page No. 8]

65. (4)

Redox reactions may involve absorption or release of heat.

[New NCERT Class 11th Page No. 9]

66. (3)

Reaction is balanced on the loss or gain of $10e^-$.

[New NCERT Class 11th Page No. 17]

67. (1)

I^- is a strong reducing agent.

[New NCERT Class 11th Page No. 13]

68. (2)

In acidic medium $MnO_4^- \rightarrow Mn^{2+}$

$n_f = 5$, Equivalent weight $= \frac{\text{Molecular mass}}{5}$

[New NCERT Class 11th Page No. 10]

69. (4)

Mole of hydrogen gas = $\frac{1}{2}$ = 0.5

Volume at NTP = mole × 22.4 L

= 0.5 × 22.4 L

= 11.2 L

[New NCERT Class 11th Page No. 21]

70. (4)

5.6 L volume at STP occupy by 11 g of gas

1 L volume at STP occupy by $\frac{11}{5.6}$ g of gas

22.4 L volume at STP occupy by $\frac{11}{5.6} \times 22.4$ g of gas

22.4 L volume at STP occupy by 44 g of gas

So, molecular mass of gas (CO_2) = 44 g

[New NCERT Class 11th Page No. 12]

71. (1)

For different atomicities of gas, number of atoms will be different.

[New NCERT Class 11th Page No. 7]

72. (1)

According to Dalton's atomic theory, atoms can neither be created nor destroyed. According to Avogadro's law, under similar condition of temperature and pressure equal volumes of all gases contain equal number of molecules.

[New NCERT Class 11th Page No. 14]

73. (1)

$H_2\overset{a}{S_2}O_7 : 2\times1 + 2a + 7\times(-2) = 0$

$\therefore$ a = +6

$K_4[\overset{b}{Fe}(CN)_6] : 4\times1 + b + 6\times(-1) = 0$

$\therefore$ b = +2

[New NCERT Class 11th Page No. 11]

74. (3)

$\because$ 1cc contains = 1.17 g

$\therefore$ 1000 cc contains = $\frac{1170}{\text{Molar mass}}$

$= \frac{1170}{36.5} = 32.05$ mol

Molarity = $\frac{32.05}{1L}$ = 32.05 mol L^{-1}

[New NCERT Class 11th Page No. 19]

75. (3)

$\because$ 100 g $CaCO_3$ contains = 6.022×10^{23} molecules

$\therefore$ 10 g $CaCO_3$ contains = $\frac{6.022\times10^{23}}{100}\times10$

$= 6.022\times10^{22}$ molecule

1 molecule of $CaCO_3$ = 50 protons

6.022×10^{22} molecule of $CaCO_3$

$= 50 \times 6.022\times10^{22}$

$= 3.011 \times 10^{24}$

[New NCERT Class 11th Page No. 5]

76. **(3)**

18 mL H_2O = 18 g H_2O = 1 mole

($\because$ d = 1g/mL and V = 18 mL)

= 6.022×10^{23} molecules

Since, 1 molecule of water contains

= $10e^-$ [$2e^-$ of H + $8e^-$ of O]

Thus, 6.022×10^{23} molecules will contain

= $10 \times 6.022 \times 10^{23} e^- = 6.022 \times 10^{24} e^-$

[New NCERT Class 11th Page No. 6]

77. **(2)**

As the molar mass of H_2O is less than that of O_2 and N_2, the molar mass of moist air is less than that of dry air.

[New NCERT Class 11th Page No. 3]

78. **(1)**

Moles of O-atoms = 9 × Moles of Cu

[New NCERT Class 11th Page No. 15]

79. **(1)**

B^{10} =10.0 × 0.196 =1.96 amu

B^{11} =11.0 × 0.804 = 8.844 amu

Total = 1.96 + 8.844 = 10.804 amu

Thus, the average atomic mass of boron is 10.804 amu.

[New NCERT Class 11th Page No. 17]

80. **(1)**

Number of gram atoms $= \dfrac{2.3}{23} = 0.1$

[Atomic mass of sodium = 23 g]

[New NCERT Class 11th Page No. 19]

81. **(1)**

Molar mass of copper = 63.5 g/mol

Number of moles in 0.635 g of copper

$= \dfrac{0.635}{63.5} = 0.01$

Number of copper atoms in one mole

= 6.022×10^{23}

Number of copper atoms in 0.01 moles

= $0.01 \times 6.022 \times 10^{23}$

= 6.022×10^{21}

[New NCERT Class 11th Page No. 20]

82. **(1)**

Σ Masses of reactants = Σ Masses of products

(3 + 11.2) g (8.8 + 5.4) g

Hence, law of conservation of mass is verified.

[New NCERT Class 11th Page No. 14]

83. **(1)**

3.4 g sulphur is present in 100 g insulin

$\therefore$ 32 g sulphur will be present in $\dfrac{100}{3.4} \times 32$ g insulin = 940

$\therefore$ Molar mass of insulin is 940 amu

[New NCERT Class 11th Page No. 10]

84. **(2)**

$$\text{Equivalent mass of element} = \frac{\text{Mass of element}}{\text{Mass of oxygen}} \times 8$$

$$= \frac{80}{20} \times 8 = 32$$

[New NCERT Class 11th Page No. 7]

85. **(1)**

Molecular mass (M) = 2 × V.D. (Vapour density)

M_1 = 2 × V.D. (1)

M_2 = 2 × V.D. (2)

Given $= \dfrac{\text{V.D.(1)}}{\text{V.D.(2)}} = \dfrac{1}{3}$

$\therefore \dfrac{M_1}{M_2} = \dfrac{1}{3}$

[New NCERT Class 11th Page No. 21]

86. **(1)**

Mass of 93 'H' atoms = 93 amu

Mass of 7 'He' atoms = 28 amu

% Hydrogen by mass $= \dfrac{93}{(93+28)} \times 100 = 77\%$

[New NCERT Class 11th Page No. 16]

87. **(1)**

Density	kg m^{3-}
Pressure	bar
Length	A°
Energy	eV

[New NCERT Class 11th Page No. 18]

88. **(1)**

Element	Percentage	Atomic mass	Relative number of atoms	Simplest ratio
Carbon	92.30	12	7.69	1
Hydrogen	7.70	1	7.70	1

Empirical formula = CH

[New NCERT Class 11th Page No. 11]

89. **(2)**

Gram atomic mass of oxygen = mass of 1 mole atom of oxygen

= (16 amu) × N_A

$= \left(\frac{16}{N_A} g\right) \times N_A$

= 16 g

[New NCERT Class 11th Page No. 19]

90. **(1)**

For A = $\frac{10}{1} = 10$; For B = $\frac{10}{5} = 2$

So, B is limiting reagent.

∵ 5 Mole of B gives 1 mole of C

∵ 10 mole of B gives 2 mole of C.

[New NCERT Class 11th Page No. 21]

91. **(2)**

Number of F^- ions = $\frac{4.2}{27 + 3 \times 19} \times 6.022 \times 10^{23} \times 3$

= 9.033×10^{22}

[New NCERT Class 11th Page No. 10]

92. **(2)**

$\overset{+5}{H N O_3} > \overset{+2}{N O} > \overset{0}{N_2} > \overset{-3}{N H_4 Cl}$

[New NCERT Class 11th Page No. 6]

93. **(4)**

H_2O_2 itself oxidized to O_2.

[New NCERT Class 11th Page No. 4]

94. **(4)**

Oxidation state of Cr in K_3CrO_8 is +5.

In CrO_5, it is +6.

K_3CrO_8 has four peroxide (O_2^{2-}) linkage while CrO_5 has only two peroxide linkage.

[New NCERT Class 11th Page No. 10]

95. **(1)**

$2Na[Ag(CN)_2] + Zn \rightarrow Na_2[Zn(CN)_4] + 2Ag$

[New NCERT Class 11th Page No. 18]

96. **(1)**

$$(Fe^{2+} \rightarrow Fe^{3+} + 1e^-) \times 6$$
$$6e^- + 14H^+ + Cr_2O_7^{2-} \rightarrow 2Cr^{3+} + 7H_2O$$
$$\overline{6Fe^{2+} + 14H^+ + Cr_2O_7^{2-} \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O}$$

$x = 6,\ y = 1,\ z = 7$

[New NCERT Class 11th Page No. 17]

97. **(3)**

Oxidizing agent must undergo reduction. I^- can never be reduced.

[New NCERT Class 11th Page No. 13]

98. **(2)**

$\overset{+7}{M}nO_4^-, \overset{+6}{Cr}O_2Cl_2$ These are highest oxidation states of Mn and Cr, respectively.

[New NCERT Class 11th Page No. 15]

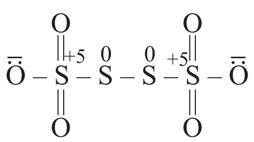
99. **(3)**

Toluene, $C_7H_8 \Rightarrow 7x + 8(+1) = 0 \Rightarrow 7x = -8$.

$x = \frac{-8}{7}$

[New NCERT Class 11th Page No. 21]

100. **(3)**



[New NCERT Class 11th Page No. 3]

SECTION-(III) BOTANY

101. (1)
The main difference between plant and animal cells is that plant cells have a cell wall on the outer layer, whereas animal cells only have a cell membrane.
[New NCERT Class 11th Page No. 88]

102. (2)
The vacuole is the membrane-bound space found in the cytoplasm. It contains water, sap, excretory product and other materials not useful for the cell. The vacuole is bound by a single membrane called tonoplast. In plant cells the vacuoles can occupy up to 90 per cent of the volume of the cell.
[New NCERT Class 11th Page No. 96]

103. (4)
In addition to the genomic DNA (the single chromosome/circular DNA), many bacteria have small circular DNA outside the genomic DNA. These smaller DNA are called plasmids.
[New NCERT Class 11th Page No. 90-91]

104. (1)
In addition to the genomic DNA (the single chromosome/circular DNA), many bacteria have small circular DNA outside the genomic DNA. These smaller DNA are called plasmid.
[New NCERT Class 11th Page No. 90]

105. (3)
An elaborate network of filamentous proteinaceous structures consisting of microtubules, microfilaments and intermediate filaments present in the cytoplasm is collectively referred to as the cytoskeleton. The cytoskeleton in a cell are involved in many functions such as mechanical support, motility, maintenance of the shape of the cell.
[New NCERT Class 11th Page No. 98]

106. (2)
Mitochondria is known as power house of the cell.It is the site of aerobic respiration.
[New NCERT Class 11th Page No. 97]

107. (3)
Cell membrane of animal cell contains lipids, proteins and carbohydrates.
[New NCERT Class 11th Page No. 93]

108. (2)
Nuclear pores are the passages through which movement of RNA and protein molecules takes place in both directions between the nucleus and the cytoplasm.
[New NCERT Class 11th Page No. 100]

109. (2)
Sometimes few chromosomes have non-staining secondary constriction at a constant location. This gives the appearance of a small fragment called the satellite.
[New NCERT Class 11th Page No. 102]

110. (2)
In some prokaryotes like cyanobacteria, there are other membranous extensions into the cytoplasm called chromatophores which contain pigments.
[New NCERT Class 11th Page No. 91]

111. (4)
The content of nucleolus is continuous with the rest of the nucleoplasm as it is not a membrane bound structure. It is a site for active ribosomal RNA synthesis.
[New NCERT Class 11th Page No. 100]

112. (1)
- Microbodies – Many membranes bound minute vesicles called microbodies that contain various enzymes, are present in both plant and animal cells.
- Cell wall and plastids are present only in plants cells.

[New NCERT Class 11th Page No. 102]

113. (1)
The metacentric chromosome has middle centromere forming two equal arms of the chromosome. The sub-metacentric chromosome has centromere slightly away from the middle of the chromosome resulting into one shorter arm and one longer arm. In case of acrocentric chromosome, the centromere is situated close to its end forming one extremely short and one very long arm. Telocentric chromosome has a terminal centromere.
[New NCERT Class 11th Page No. 101]

114. (2)
A special membranous structure is the mesosoma which is formed by the extensions of plasma membrane into the cell.
[New NCERT Class 11th Page No. 90]

115. (3)

The axoneme usually has nine doublets of radially arranged peripheral microtubules, and a pair of centrally located microtubules. Such an arrangement of axonemal microtubules is referred to as the 9+2 array.

[New NCERT Class 11th Page No. 99]

116. (4)

Cell membranes are involves in active transport of materials.

[New NCERT Class 11th Page No. 93-94]

117. (2)

The isolated lysosomal vesicles have been found to be very rich in almost all types of hydrolytic enzymes (hydrolases – lipases, proteases, carbohydrases) optimally active at the acidic pH. These enzymes are capable of digesting carbohydrates, proteins, lipids and nucleic acids.

[New NCERT Class 11th Page No. 96]

118. (4)

In prokaryotes, the genetic material is not enclosed in a well-defined nucleus.

[New NCERT Class 11th Page No. 90]

119. (3)

A number of proteins synthesised by ribosomes on the endoplasmic reticulum are modified in the cisternae of the golgi apparatus before they are released from its trans face or concave face.

[New NCERT Class 11th Page No. 96]

120. (2)

In the chromoplasts fat soluble carotenoid pigments like carotene, xanthophylls and others are present. This gives the part of the plant a yellow, orange or red colour.

[New NCERT Class 11th Page No. 97]

121. (1)

Some mature cells even lack nucleus, e.g., erythrocytes of many mammals and sieve tube cells of vascular plants.

[New NCERT Class 11th Page No. 100]

122. (3)

Thylakoids are present in stroma of chloroplast.

[New NCERT Class 11th Page No. 98]

123. (4)

Every chromosome (visible only in dividing cells) essentially has a primary constriction or the centromere on the sides of which disc shaped structures called kinetochores are present.

[New NCERT Class 11th Page No. 101]

124. (2)

The endomembrane system includes endoplasmic reticulum (ER), golgi complex, lysosomes and vacuoles. Since the functions of the mitochondria, chloroplast and peroxisomes are not coordinated with the above components, these are not considered as part of the endomembrane system.

[New NCERT Class 11th Page No. 95]

125. (1)

The two subunits of 80S ribosomes are 60S and 40S while that of 70S ribosomes are 50S and 30S.

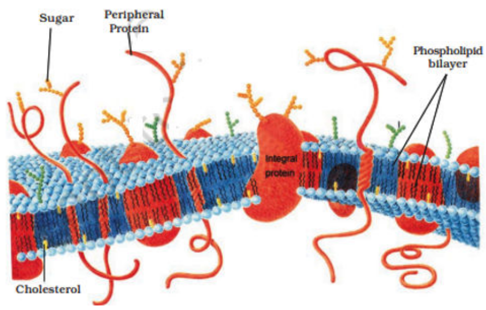
[New NCERT Class 11th Page No. 98]

126. (4)

- A represent outer membrane – Forms the continuous limiting boundary of the organelle.
- B represent cristae – Increases surface area.
- C represent mitochondria matrix – a possesses single circular DNA molecule, a few RNA molecules, ribosomes (70S) and the components required for the synthesis of proteins.

[New NCERT Class 11th Page No. 97]

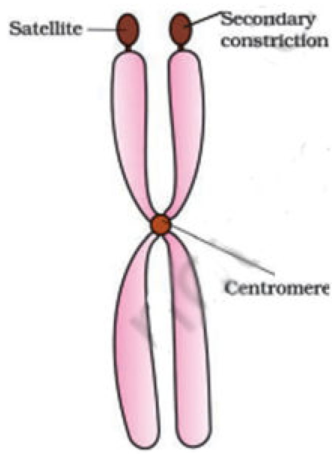
127. (3)



A-Sugar	B-Peripheral protein	C-Phospholipid bilayer	D-Integral protein

[New NCERT Class 11th Page No. 93]

128. (4)



[New NCERT Class 11th Page No. 101]

129. (2)

Bacillus	Rod shaped
Coccus	Spherical shaped
Vibrio	Comma shaped
Spirillum	Spiral shaped

[New NCERT Class 11th Page No. 89]

130. (3)

Inclusion bodies	Reserve materials in the cytoplasm of prokaryotes
Plasmid DNA	Confers certain unique phenotypic characters to bacteria
Polysome	Several ribosomes attach to a single mRNA
Mesosome	Increases surface area and enzymatic content of plasma membrane

[New NCERT Class 11th Page No. 90-91]

131. (2)

- Enzymes of lysosomes are optimally active at acidic pH, not basic pH.
- ER are actively involved in protein synthesis and also the major site for synthesis of lipid.

[New NCERT Class 11th Page No. 95 & 96]

132. (3)

Typical bacteria	1 – 2 μm
Viruses	0.02 μm
PPLO	0.1 μm
Typical eukaryotic cell	10 – 20 μm

[New NCERT Class 11th Page No. 90]

133. (1)

Anton Von Leeuwenhoek	First saw and described a live cell
Rudolf Virchow	Give final shape to the cell theory
Robert Brown	First described nucleus as a cell organelle
Singer and Nicolson	Proposed fluid mosaic model of plasma membrane

[New NCERT Class 11th Page No. 87,88, 93]

134. (1)

The Golgi apparatus is a membrane-bound organelle, not non-membrane-bound.

[New NCERT Class 11th Page No. 88, 94]

135. (3)

Material of the nucleus stained by the basic dyes	Chromatin
Pericentriolar material	Centrosome
Arise from the centriole-like structure	Cilia and flagella
Contractile vacuoles	Osmoregulation

[New NCERT Class 11th Page No. 96,99, 100,101]

136. (2)

Rudolf Virchow (1855) first explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e cellula*). He modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape. Cell theory as understood today is:

(i) All living organisms are composed of cells and products of cells.

(ii) All cells arise from pre-existing cells.

[New NCERT Class 11th Page No. 88]

137. (1)

All prokaryotes have a cell wall surrounding the cell membrane except in mycoplasma.

[New NCERT Class 11th Page No. 90]

138. (2)

The matrix of mitochondria contains the components required for the synthesis of proteins.

[New NCERT Class 11th Page No. 96]

139. (4)

- Mitochondria unless specifically stained, are not easily visible under the microscope.
- Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis.
- Gas vacuoles are inclusion bodies.

[New NCERT Class 11th Page No. 91,96,100]

140. (3)

- The leucoplasts are the colourless plastids.
- The functions of the mitochondria, chloroplast and peroxisomes are not co-ordinated.

[New NCERT Class 11th Page No. 94, 95, 96, 97]

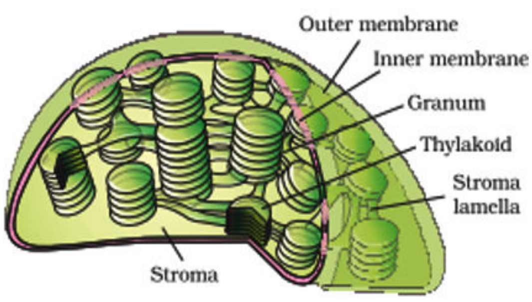
141. (4)

RER are extensive and continuous with the outer membrane of the nucleus.

[New NCERT Class 11th Page No. 95]

142. (4)

A-Outer membrane; B-Grana; C-Stroma



[New NCERT Class 11th Page No. 98]

143. (3)

- The chloroplasts contain chlorophyll and carotenoid pigments which are responsible for trapping light energy essential for photosynthesis.
- In the chromoplasts fat soluble carotenoid pigments like carotene, xanthophylls and others are present.

[New NCERT Class 11th Page No. 97- 99]

144. (3)

The detailed structure of the membrane was studied only after the advent of the electron microscope in the 1950s.

[New NCERT Class 11th Page No. 91,93]

145. (3)

In bacteria, flagella primarily help in motility (movement) but do not have a role in capturing food.

[New NCERT Class 11th Page No. 89, 91, 95,98]

146. (3)

The content of nucleolus is continuous with the rest of the nucleoplasm as it is not a membrane bound structure

[New NCERT Class 11th Page No. 100]

147. (2)

The cell wall of a young plant cell, the primary wall is capable of growth, which gradually diminishes as the cell matures and the secondary wall is formed on the inner (towards membrane) side of the cell.

[New NCERT Class 11th Page No. 94]

148. (4)

R is not the correct explanation of A because the stroma refers to the internal fluid-filled space, while the stroma lamellae are membranous structures connecting the thylakoids of the different grana.

[New NCERT Class 11th Page No. 98]

149. (2)

Ribosomes are the granular structures.

[New NCERT Class 11th Page No. 98]

150. (4)

The R is not the correct explanation of A, because the visibility of chromosomes during cell division is due to DNA condensation, not because of the length of the DNA.

[New NCERT Class 11th Page No. 101]

SECTION-(IV) ZOOLOGY

151. (3)

Ligaments attach one bone to another, is an example of dense regular connective tissue.

[Old NCERT Class 11th Page No. 103]

152. (4)

Epithelial tissue is distinguished from connective, muscular, or nervous tissue by its ability to act as lining or covering over body cavities, ducts and tubes.

[Old NCERT Class 11th Page No. 101]

153. (4)

Cockroaches are dioecious and both sexes have well developed reproductive organs.

[Old NCERT Class 11th Page No. 114]

154. (1)

- Squamous epithelium lines the alveoli of lungs and endothelium.
- Compound epithelium lines the dry surface of skin, inner lining of ducts of salivary glands.
- Glandular epithelium lines the goblet cells.

[Old NCERT Class 11th Page No. 101, 102]

155. (1)

- The nictitating membrane protects the eyes of the frog in water.
- The tympanum is the ear of the frog.
- In frogs, cloaca is a small median chamber used to pass faecal matter, urine and gametes to exterior.
- Bidder's canal is a part of the male reproductive system of frogs.

[Old NCERT Class 11th Page No. 116]

156. (3)

Each forelimb of frog has four digits while each hindlimb of frog has five digits.

[Old NCERT Class 11th Page No. 116]

157. (3)

Sclerite	Exoskeleton plate
Tegmina	Forewings
Gizzard	Proventriculus
Anal cerci	10th segment

[Old NCERT Class 11th Page No. 113]

158. (4)

Forewings of cockroaches arises from mesothorax.

[Old NCERT Class 11th Page No. 112]

159. (1)

Vasa efferentia in frog are 10-12 in number that arise from testes.

[Old NCERT Class 11th Page No. 119]

160. (3)

- Muscles are of three types; skeletal, smooth, and cardiac muscles.
- Endocrine glands secrete their products in the fluid surrounding the tissue and they do not have ducts.
- Muscles play an active role in all the movements of the body.
- Cuboidal epithelium has cube-like cells.

[Old NCERT Class 11th Page No. 104]

161. (1)

- Head of cockroach is formed by fusion of six segment.
- The abdomen in both males and females consists of 10 segments.
- Each thoracic segment bears a pairs of legs.

[Old NCERT Class 11th Page No. 112]

162. (3)

The hindgut of the cockroach is divided into three parts, namely, the ileum, the colon and the rectum. Duodenum is not a structure of the hindgut.

[Old NCERT Class 11th Page No. 113]

163. (4)

Smooth muscles are present in wall of internal organs such as the blood vessels, stomach and intestine.

[Old NCERT Class 11th Page No. 101]

164. (3)

- The forelimbs and hindlimbs of frogs help in swimming, walking, leaping and burrowing.
- Body of a frog is divisible into head and trunk.

[Old NCERT Class 11th Page No. 116]

165. (3)

There are ten pairs of cranial nerves arising from the brain of frog.

[Old NCERT Class 11th Page No. 119]

166. (2)

Bones have a hard and non-pliable ground substance rich in calcium salts and collagen fibres which give bone its strength. They interact with skeletal muscles attached to them to bring about movements.

[Old NCERT Class 11th Page No. 104]

167. (1)

The mouth-parts of the cockroach consist of a labrum (upper lip), a pair of mandibles, a pair of maxillae and a labium (lower lip). A median flexible lobe acts as a tongue, which is also known as hypopharynx.

[Old NCERT Class 11th Page No. 112]

168. (2)

- Compound epithelium is made of more than one layer (multi-layered) of cells and thus has a limited role in secretion and absorption.
- Columnar epithelium is found in the lining of stomach and intestine.

[Old NCERT Class 11th Page No. 101, 102]

169. (3)

- *Rana tigrina* or frog possesses a bilobed tongue to capture food.
- It has webbed feet for the purpose of swimming in water.
- It also has a membranous tympanum to receive sound signals.
- It has moist, slippery skin.

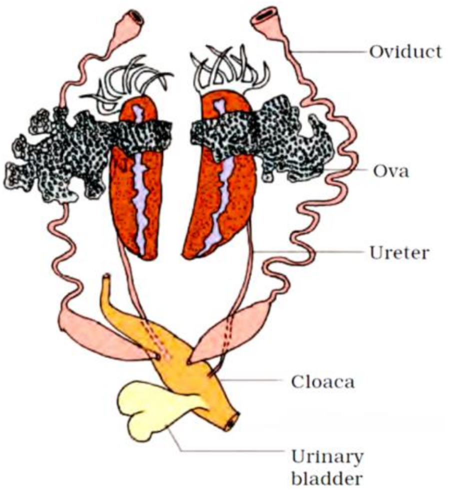
[Old NCERT Class 11th Page No. 116]

170. (2)

The first pair of wings arises from mesothorax and the second pair from metathorax. The hind wings are transparent, membranous and are used in flight.

[Old NCERT Class 11th Page No. 112]

171. (3)



[Old NCERT Class 11th Page No. 119]

172. (3)

- Midbrain of frogs is characterised by a pair of optic lobes.
- Hindbrain of frogs consists of cerebellum and medulla oblongata.

[Old NCERT Class 11th Page No. 119]

173. (1)

Head of cockroach is triangular in shape and lies anteriorly at right angles to the longitudinal body axis. It is formed by the fusion of six segments and shows great mobility in all directions due to its flexible neck.

[Old NCERT Class 11th Page No. 112]

174. (4)

Adipose tissue	Fat storage
Dense irregular connective tissue	Skin
Fluid connective tissue	Blood
Areolar tissue	Macrophage

[Old NCERT Class 11th Page No. 103]

175. (2)

In male cockroaches, the external genitalia are represented by male gonapophysis or phallomere (chitinous asymmetrical structures, surrounding the male gonopore).

[Old NCERT Class 11th Page No. 114]

176. (3)

- The lungs of frogs are a pair of elongated, pink coloured sac-like structures.
- Oviducts are not present in male frogs.
- They have both hepatic and renal portal systems and sexual dimorphism is present in frogs.

[Old NCERT Class 11th Page No. 118]

177. (2)

Cockroaches have mosaic vision with more sensitivity but less resolution, being common during night (hence called nocturnal vision).

[Old NCERT Class 11th Page No. 114]

178. (4)

A ring of 6-8 blind tubules called hepatic or gastric caeca is present at the junction of foregut and midgut, which secrete digestive juice.

[Old NCERT Class 11th Page No. 113]

179. (1)

- Blood vessels - Smooth muscle fibres
- Heart wall - Cardiac muscle fibres
- Biceps - Skeletal muscle fibres
- Muscles in intestine - Smooth muscles fibres

[Old NCERT Class 11th Page No. 105]

180. (1)

Neural tissue exerts the greatest control over the body's responsiveness to changing conditions.

[Old NCERT Class 11th Page No. 104]

181. (2)

When two or more organs perform a common function by their physical/chemical interaction, they together form an organ system.

[Old NCERT Class 11th Page No. 100]

182. (3)

- Frog RBCs are nucleated.
- In unicellular organisms all functions like digestion, respiration and reproduction are performed by a single cell. In complex multicellular animals, the same basic functions are carried out by different groups of cells in a well-organised manner.

[Old NCERT Class 11th Page No. 100]

183. (1)

- Hormone is the secretion of endocrine glands.
- Saliva, earwax, mucus and digestive enzymes are secretions of exocrine gland.

[Old NCERT Class 11th Page No. 102]

184. (3)

- Frogs use camouflage.
- Frogs have the ability to change the colour to hide from their enemies.

[Old NCERT Class 11th Page No. 116]

185. (4)

Neuroglia are non-neuronal cells that support and protect the neurons.

[Old NCERT Class 11th Page No. 105]

186. (3)

Tendons and ligaments are the examples of dense regular connective tissue.

[Old NCERT Class 11th Page No. 103]

187. (4)

Bone is the main tissue that provides structural frame to the body. Bones support and protect softer tissues and organs. The bone marrow in some bones is the site of production of blood cells.

[Old NCERT Class 11th Page No. 104]

188. (1)

- In frogs, forebrain includes olfactory lobes, paired cerebral hemispheres and unpaired diencephalon.
- The midbrain is characterised by a pair of optic lobes.
- Hindbrain consists of cerebellum and medulla oblongata.

[Old NCERT Class 11th Page No. 118]

189. (4)

Compound epithelium covers the dry surface of the skin, the moist surface of buccal cavity, pharynx, inner lining of ducts of salivary glands and of pancreatic ducts.

[Old NCERT Class 11th Page No. 102]

190. (4)

The nymphs of cockroaches grow by moulting about 13 times to reach the adult form.

[Old NCERT Class 11th Page No. 115]

191. (1)

In female cockroach, the 7th sternum is boat shaped.

[Old NCERT Class 11th Page No. 114]

192. (4)

Frogs respire on land and in the water by two different methods. In water, skin acts as aquatic respiratory organ (cutaneous respiration). Dissolved oxygen in the water is exchanged through the skin by diffusion.

[Old NCERT Class 11th Page No. 117]

193. (4)

Cartilage	Between adjacent bones of limbs
Cube-like cells	Ducts of glands
Tall and slender cells	Stomach
Flattened cells	Air sacs of lungs

[Old NCERT Class 11th Page No. 102]

194. (3)

- Adhering junctions perform cementing to keep neighbouring cells together.
- Gap junctions facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules.
- Tight junctions help to stop substances from leaking across a tissue.

[Old NCERT Class 11th Page No. 102]

195. **(4)**

The intercellular material of cartilage is solid and pliable and resists compression. Cartilage is present in the tip of nose between adjacent bones of the vertebral column, limbs and hands in adults.

[Old NCERT Class 11th Page No. 102]

196. **(2)**

The given figure is of areolar tissue which is a type of loose connective tissue and "X" refers to the fibroblast.

[Old NCERT Class 11th Page No. 103]

197. **(3)**

The compound eyes in cockroaches consist of 2000 hexagonal ommatidia.

[Old NCERT Class 11th Page No. 114]

198. **(4)**

Loose connective tissue has cells and fibres loosely arranged in a semi-fluid ground substance.

[Old NCERT Class 11th Page No. 103]

199. **(3)**

In cockroaches, the respiratory system consists of a network of trachea that opens through 10 pairs of small holes called spiracles which are present on the lateral side of the body.

[Old NCERT Class 11th Page No. 113, 114]

200. **(3)**

Frogs help to maintain ecological balance. The frogs serve as an important link in the food chain and food web.

[Old NCERT Class 11th Page No. 120]